Coffee Shop Sales Analysis Report Using Excel

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Introduction

This report provides a detailed analysis of coffee shop sales data sourced from Kaggle, covering transactions, product sales, customer footfall, and revenue trends. The dataset includes 149,117 records and 18 columns, supplemented with additional derived columns for detailed insights. The analysis aims to uncover sales patterns across days, hours, months, store locations, and product categories.

To better understand real-world scenarios, this report includes case-study-style explanations addressing potential business challenges and opportunities.

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Data Overview

Total Sales Revenue: $698,812.33

Total Footfall: 149,116 customers

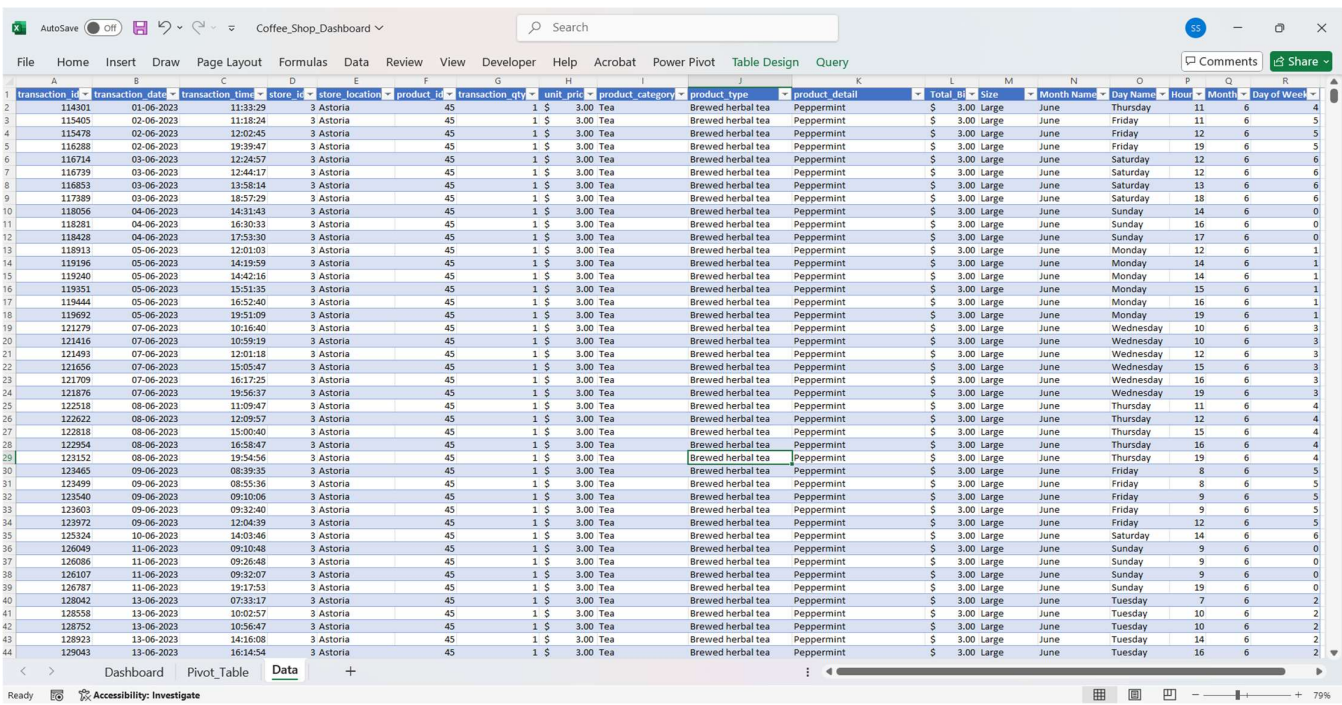
Average Bill per Person: $4.69 Average

Orders per Person: 1.44

Columns Added:

- Month Name, Day Name, Hour (extracted from transaction_date and transaction_time)
- Total Bill (calculated using transaction_qty and unit_price)

Analysis Tools Used: Power Query Editor, Pivot Tables and Excel Dashboard



transaction_id	transaction_date	transaction_time	store_id	store_location	product_id	transaction_qty	unit_pric	product_category	product_type	product_detail	Total_Bi	Size	Month Name	Day Name	Hour	Month	Day of Week
114301	01-06-2023	11:33:29	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Thursday	11	6	4
115405	02-06-2023	11:18:24	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Friday	11	6	5
115478	02-06-2023	12:02:45	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Friday	12	6	5
116288	02-06-2023	19:39:47	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Friday	19	6	5
116714	03-06-2023	12:24:57	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Saturday	12	6	6
116739	03-06-2023	12:44:17	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Saturday	12	6	6
116853	03-06-2023	13:58:14	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Saturday	13	6	6
117389	03-06-2023	18:57:29	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Saturday	18	6	6
118056	04-06-2023	14:31:43	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Sunday	14	6	0
118281	04-06-2023	16:30:33	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Sunday	16	6	0
118428	04-06-2023	17:53:30	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Sunday	17	6	0
118913	05-06-2023	12:01:03	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Monday	12	6	1
119196	05-06-2023	14:19:59	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Monday	14	6	1
119240	05-06-2023	14:42:16	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Monday	14	6	1
119351	05-06-2023	15:51:35	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Monday	15	6	1
119444	05-06-2023	16:52:40	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Monday	16	6	1
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122818	08-06-2023	15:00:40	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Thursday	15	6	4
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123465	09-06-2023	08:39:35	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Friday	8	6	5
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123540	09-06-2023	09:10:06	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Friday	9	6	5
123603	09-06-2023	09:32:40	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Friday	9	6	5
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125324	10-06-2023	14:03:46	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Saturday	14	6	6
126049	11-06-2023	09:10:48	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Sunday	9	6	0
126086	11-06-2023	09:26:48	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Sunday	9	6	0
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128558	13-06-2023	10:02:57	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Tuesday	10	6	2
128752	13-06-2023	10:56:47	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Tuesday	10	6	2
128923	13-06-2023	14:16:08	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Tuesday	14	6	2
129043	13-06-2023	16:14:54	3	Astoria	45	1	$ 3.00	Tea	Brewed herbal tea	Peppermint	$ 3.00	Large	June	Tuesday	16	6	2

Key Insights and Analys

1. Sales Variation by Day of Week and Hour

Case Scenario: A coffee shop owner notices long queues during specific times of the day, which impacts customer satisfaction and leads to lost sales opportunities during peak hours.

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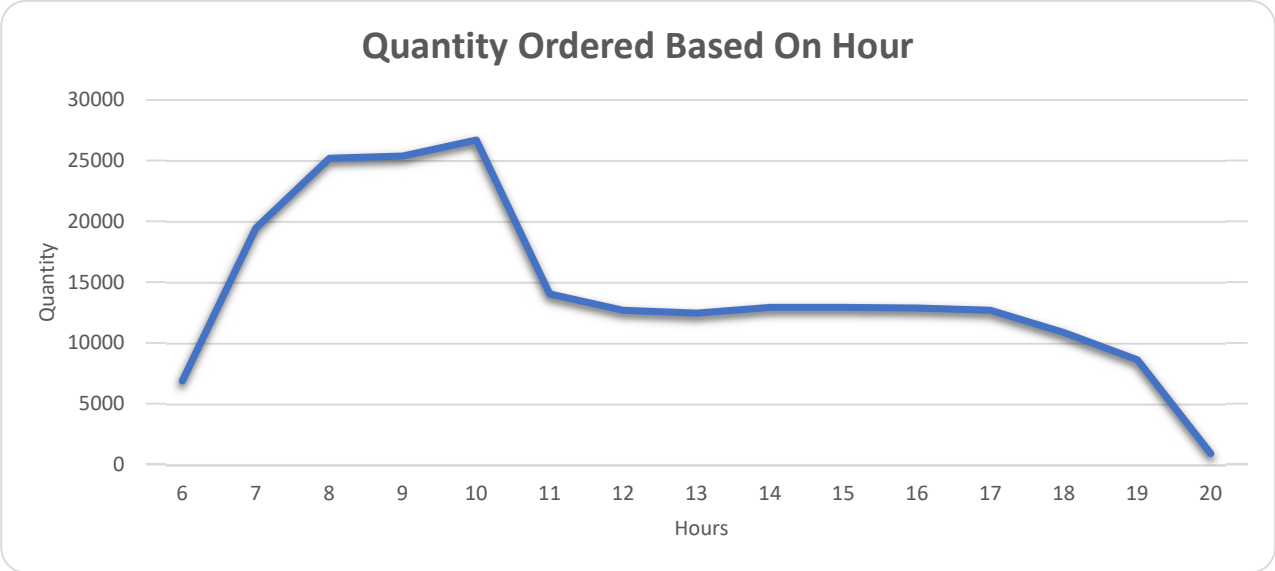
Peak Hours: Sales are highest between 8:00 AM and 10:00 AM, with transaction quantities peaking at 26,713 at 10:00 AM.

Off-Peak Hours: Sales decline sharply after 12:00 PM and are minimal by 8:00 PM.

Footfall Patterns: The highest footfall occurs during mornings, indicating strong demand for breakfast items and beverages.

Row Lab	Sum of Total_Bill
January	$81,677.74
February	$76,145.19
March	$98,834.68
April	$1,18,941.08
May	$1,56,727.76
June	$1,66,485.88
Grand Tota	**$6,98,812.33**

Row Labels	Sum of transaction_qty
10	26713
9	25370
8	25197
7	19449
11	14035
15	12923
14	12907
16	12881
17	12700
12	12690
13	12439
18	10826
19	8595
6	6865
20	880
Grand Total	**214470**



Recommendations:

- Optimize staffing during peak hours (6:00 AM - 11:00 AM) to reduce wait times.
- Offer promotions or discounts during off-peak hours to increase sales and utilize idle capacity.
- Introduce pre-order options to streamline operations during busy hours.

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2. Monthly Sales Revenue Trends

Case Scenario: A marketing team wants to plan targeted campaigns to boost sales during low-performing months.

Insights: Sales increase consistently from January to June, with May and June contributing the highest revenue.

Seasonality: The upward trend could indicate increased customer spending during spring and early summer months.

Row Labels	Count of transaction_id	Sum of Total_Bill
Sunday	21096	$98,330.31
Monday	21643	$1,01,677.28
Tuesday	21202	$99,455.94
Wednesday	21310	$1,00,313.54
Thursday	21654	$1,00,767.78
Friday	21701	$1,01,373.00
Saturday	20510	$96,894.48
Grand Total	**149116**	**$6,98,812.33**

Recommendations:

- Introduce seasonal products or promotions during high-revenue months to maximize profits.
- Offer loyalty rewards or discounts during January and February to encourage more spending during these slower months.

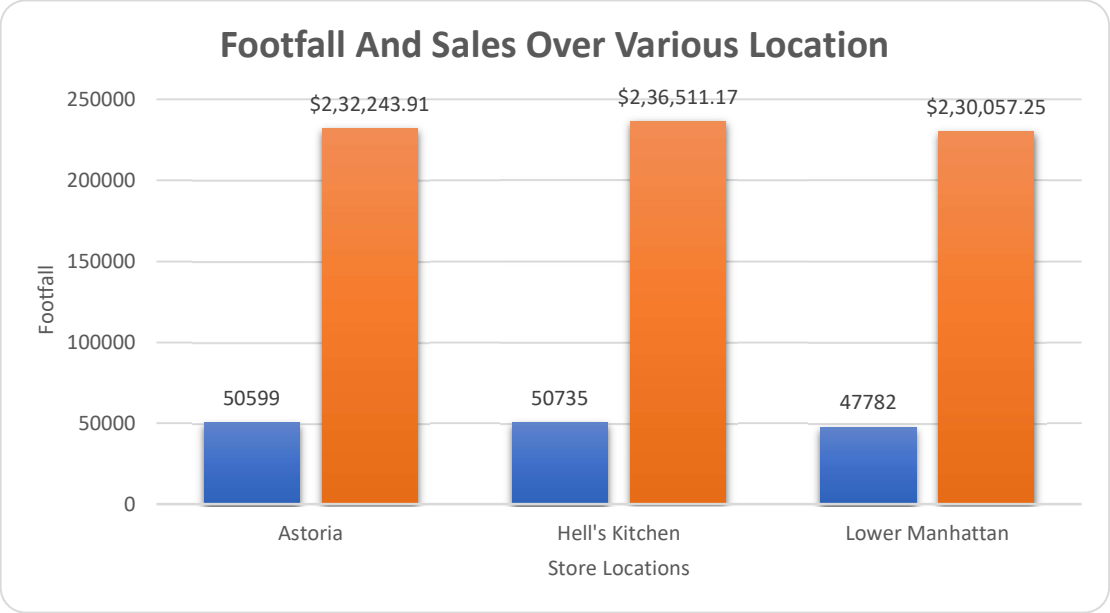
3. Sales Variation Across Store Locations

Case Scenario: Management wants to identify why one store location outperforms others and how to replicate this success in underperforming areas.

Insights: Hell's Kitchen leads in both footfall and total sales revenue, followed closely by Astoria and Lower Manhattan.

Performance: All three locations demonstrate strong sales and similar revenue generation.

Row Labels	Count of transaction_id	Sum of Total_Bill
Astoria	50599	$2,32,243.91
Hell's Kitchen	50735	$2,36,511.17
Lower Manhattan	47782	$2,30,057.25
Grand Total	**149116**	**$6,98,812.33**



Recommendations:

- Investigate customer preferences and demographics in Hell's Kitchen to replicate success in other locations.
- Promote unique offerings or localized campaigns in Lower Manhattan to increase footfall.
- Implement feedback systems to understand location-specific customer needs.

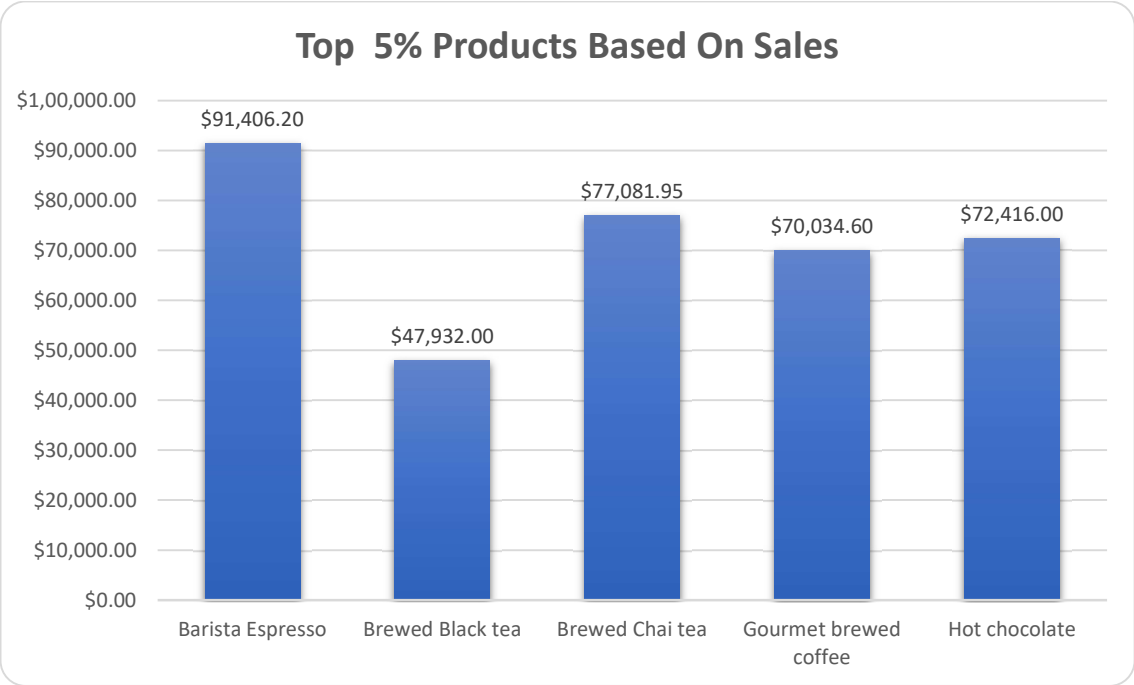
4. Best-Selling Products by Revenue

Case Scenario: A product manager seeks to identify top-performing items to prioritize inventory and marketing efforts.

Insights: Barista Espresso is the top-performing product, contributing significantly to overall revenue.

Product Preferences: Hot beverages dominate the top 5 products, reflecting customer demand for warm drinks.

Row Labels	Sum of Total_Bill
Barista Espresso	$91,406.20
Brewed Black tea	$47,932.00
Brewed Chai tea	$77,081.95
Gourmet brewed cofl	$70,034.60
Hot chocolate	$72,416.00
Grand Total	**$3,58,870.75**



Recommendations:

- Highlight and promote best-selling products in-store and online.
- Bundle top products with complementary items to increase average order value.
- Use customer feedback to refine or enhance the recipes of top-selling products.

5. Sales by Product Category

Case Scenario: A business development team evaluates the contribution of each product category to decide where to focus future investments.

Insights: Coffee and tea categories contribute 67% of total revenue, with bakery items being the third-largest revenue generator.

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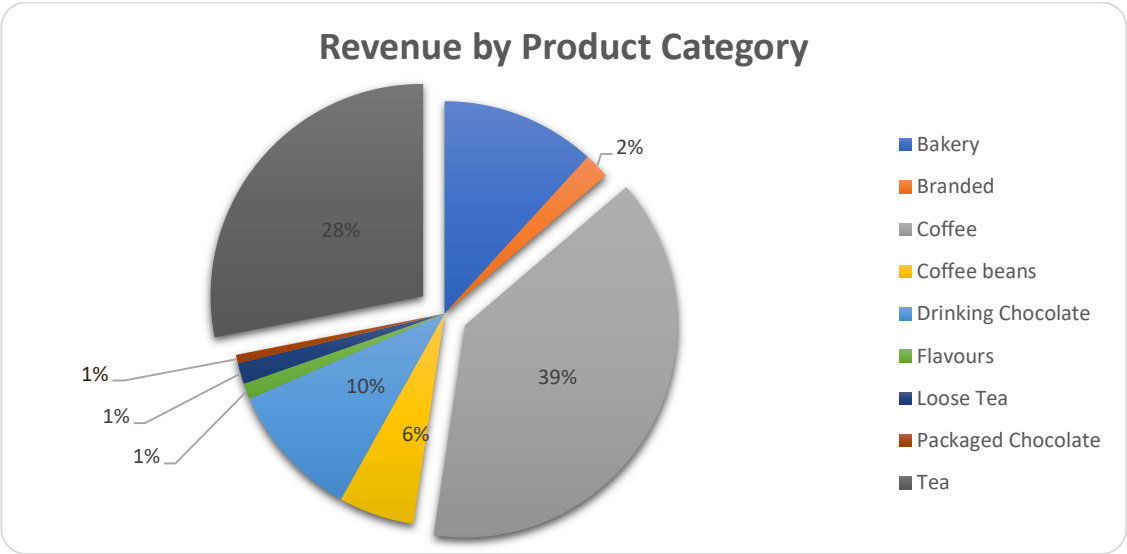
Row Labels	Sum of Total_Bill
Bakery	$82,315.64
Branded	$13,607.00
Coffee	$2,69,952.45
Coffee beans	$40,085.25
Drinking Chocolate	$72,416.00
Flavours	$8,408.80
Loose Tea	$11,213.60
Packaged Chocolate	$4,407.64
Tea	$1,96,405.95
Grand Total	**$6,98,812.33**



Recommendations:

- Expand coffee and tea product lines to cater to customer preferences.
- Offer combo deals with bakery items to boost cross-category sales.
- Conduct market research to explore emerging trends in beverages and bakery items.

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Additional Insights

Average Order Size:

The average order value is relatively low ($1.44). Encouraging larger orders through upselling and bundled deals could significantly increase revenue.

Customer Preferences by Size:

Regular and large-sized orders contribute nearly 61% of total sales, suggesting that customers prefer value-for-money options.

Customer Retention Opportunities:

Leveraging loyalty programs and personalized offers can improve customer retention and increase repeat business.

Conclusion

The analysis highlights consistent growth in sales over months, strong performance during morning hours, and a preference for coffee and tea products. By optimizing operations during peak hours, tailoring promotions for off-peak times, and focusing on best-selling products, the coffee shop can further enhance revenue and customer satisfaction. Implementing these strategies, coupled with effective use of visual dashboards, will ensure sustained growth and profitability.

This report serves as a roadmap for addressing real-world challenges, such as managing peak-hour demand, improving low-performing months, and tailoring offerings to customer preferences. By taking a data-driven approach, the coffee shop can position itself for longterm success.

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